

KADEK ANDRA WIKANJAYA PUTRA

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Informatics student at Institut Teknologi Sepuluh Nopember with 2 years of hands-on experience as an AI Engineer, specializing in building end-to-end intelligent solutions. Skilled in architecting RAG workflows, LLM orchestration, and machine learning pipelines with a deep focus on complex data analysis and system optimization. Combines a rigorous technical foundation with strong interests in strategic thinking and managerial development to deliver impactful AI-driven solutions across diverse industries.

Education

Institut Teknologi Sepuluh Nopember (ITS) - Surabaya, Indonesia

Aug 2024 – Present

Bachelor of Informatics Engineering – GPA: 3.74 / 4.00

Related Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Artificial Intelligence, Machine Learning, Operating Systems, Computer Networks, Discrete Mathematics.

Work Experiences

Information Intelligent Management - Surabaya, Indonesia

Sept 2026 – Present

Research Assistant

- Designing a closed-loop agentic AI architecture using LangGraph to concurrently process multi-parameter IoT sensor streams and optical camera feeds for microalgae bioremediation in POME.
- Developing an adaptive visual preprocessing pipeline to handle high turbidity and dark media coloration for real-time monitoring of biomass condition, foaming, and flocculation.
- Formulating a domain-specific RAG knowledge base using PostgreSQL (pgvector) to retrieve cultivation SOPs and operational guidelines for automated anomaly diagnosis.
- Establishing deterministic validation guardrails using Pydantic to implement self-correction loops, preventing model hallucination and ensuring proposed actuator controls stay within safe physical thresholds before hardware execution.

Organizational Experience

UKM Musik ITS - Surabaya, Indonesia

General Secretary

Feb 2026 – Present

- Designed an end-to-end correspondence workflow and tracking system—from initial drafting to final approval—leveraging spreadsheet automation, improving administrative turnaround efficiency by 60%.
- Conducted training workshops for 50+ staff members to elevate organizational literacy regarding standardized secretarial protocols and official documentation workflows.
- Synthesized staff evaluation summaries and operational feedback across all divisions, coordinating directly with the President/Chairperson to support strategic decision-making.

Staff of Human Resources

Mar 2025 – Jan 2026

- Orchestrated routine practice schedules and real-time monitoring via centralized spreadsheets for 100+ musicians and band members across multiple rehearsal sessions.

- Facilitated one-on-one and group consultation sessions to identify internal group friction, resolve scheduling conflicts, and support band preparation.
- Documented live band performances and rehearsal milestones to archive organizational assets and assess overall stage readiness.

AIESEC Future Leaders - Surabaya, Indonesia

Project Manager

May 2025 – Jun 2025

- Spearheaded the planning and execution of a capstone business case project titled "Noona," conceptualizing a specialized herbal tea brand designed to relieve menstrual pain (dysmenorrhea).
- Guided product positioning, value proposition, and social mission, directly aiming to generate tangible health awareness and improve physical well-being for women.
- Served as the central strategic bridge between internal team members and external stakeholders in a business simulation environment, coordinating workflow milestones, pitch materials, and deliverable defense.

Delegate

Apr 2025 – Jun 2025

- Actively engaged in intensive networking and team-building sessions, establishing strong interpersonal chemistry, mutual trust, and effective team dynamics within a newly formed collaborative cohort.
- Expanded professional connections across diverse backgrounds while mastering core leadership competencies, communication protocols, and strategic problem-solving techniques.
- Deepened understanding of the United Nations Sustainable Development Goals (SDGs), focusing on human-centric impact frameworks to address real-world socio-economic and community challenges.

Himpunan Mahasiswa Teknik Computer-Informatika ITS - Surabaya, Indonesia

Vice Project Officer of Graduation Ceremony 134 HMTc ITS

May 2026 – Present

- Directed and supervised a committee of 50+ staff across diverse divisions to ensure operational alignment, seamless communication, and milestone execution.
- Coordinated crowd-handling protocols, event flow, and logistical readiness for 300+ graduates, driving a smooth and timely ceremonial ceremony.

Head of Publication and Documentation of Graduation Ceremony 133 HMTc ITS

Mar 2026 – Apr 2026

- Standardized design workflows and cross-divisional request SOPs, systematically coordinating staff from visual conceptualization and design execution to comprehensive evaluations.
- Served as the primary event photographer and managed digital media assets, cataloging and archiving post-event visual documentation.

Staff of Internal Affairs

Mar 2026 – Present

- Strengthened peer cohesion and a sense of belonging among informatics students through targeted internal engagement initiatives and structured bonding activities.
- Spearheaded organizational benchmarking sessions with external student associations to exchange organizational knowledge, best practices, and collaborative insights.

Schematics ITS - Surabaya, Indonesia

Expert Staff of Data Coordination Officer

May 2026 – Present

- Architected and managed centralized databases to process, validate, and structure records for 1,000+ national competition participants across multiple sub-events.
- Served as the primary liaison between participants and the committee, handling payment verifications, group onboarding, administrative onboarding, and registration data discrepancies.
- Collaborated closely with cross-functional event divisions to maintain seamless data synchronization, tracking pipelines, and transparent communication flows.
- Provided prompt internal technical support and guidance to committee staff, resolving operational bottlenecks, inquiries, and data anomalies.

Projects

Asah Led by Dicoding in Association with Accenture - Remote

Aug 2025 – Jan 2026

Machine Learning Cohort Awardee

- Awarded a fully funded national scholarship bootcamp valued at IDR 14,000,000, organized by Dicoding Indonesia in collaboration with Accenture.
- Mastered comprehensive machine learning engineering competencies, spanning data preprocessing, exploratory data analysis, predictive modeling, and model evaluation techniques.
- Honed professional soft skills and project management principles, including agile workflows, cross-functional collaboration, and strategic problem-solving for tech-driven solutions.

MIRR-Learn (Mirror Learning Application)

Nov 2025 – Dec 2025

- Architected an AI-powered reflective learning application as a capstone project, dynamically tailoring individualized study roadmaps based on user progress and self-reflection metrics.
- Integrated adaptive recommendation algorithms to analyze learner comprehension gaps and synthesize real-time educational pathways.
- Built an intuitive end-to-end interface and data pipeline to track long-term skill acquisition and deliver personalized feedback loops for continuous mastery.

Personal Projects - Remote

Jul 2026 – Present

Interactive Web Portfolio

- Designed and deployed a responsive personal portfolio featuring a modern, minimalist UI to effectively showcase software engineering and machine learning projects.
- Optimized asset delivery, client-side rendering, and layout aesthetics to ensure seamless navigation, high accessibility, and high performance across all devices.

AIAssistant (RAG-Powered Intelligent Assistant)

- Architected an intelligent conversational assistant utilizing Retrieval-Augmented Generation (RAG) to query, synthesize, and extract knowledge from custom document repositories.
- Implemented a vector database pipeline with semantic embedding search, enabling contextual retrieval while suppressing hallucination during grounded dialogue generation.

Intelligent CV Summarizer

- Built an automated resume parsing tool powered by Large Language Models (LLMs) to ingest unstructured CV documents and extract key competencies, experience, and education.
- Engineered structured prompt templates and JSON output formatting to deliver concise executive summaries and role-fit analyses for recruiters.

Skills

Hard Skills: Python, Machine Learning, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Data Analysis, SQL, Git

Soft Skills: Analytical Problem Solving, Strategic Leadership, Cross-Functional Collaboration, Project Management, Adaptability

Certifications

Building Generative AI Applications with Microsoft Azure – Dicoding Indonesia

Jan 2026

Credential ID: N9ZO2860RPG5

Prompt Engineering for Software Developers – <i>Dicoding Indonesia</i> Credential ID: 0LZ05LLONX65	<i>Jan 2026</i>
Applied Machine Learning – <i>Dicoding Indonesia</i> Credential ID: 4EXG3DKEDZRL	<i>Jan 2026</i>
Building Machine Learning Systems – <i>Dicoding Indonesia</i> Credential ID: 53XEK52QKXRN	<i>Dec 2025</i>
Fundamentals of Deep Learning – <i>Dicoding Indonesia</i> Credential ID: 1RXYQ4G0QZVM	<i>Aug 2025</i>
DevOps Fundamentals – <i>Dicoding Indonesia</i> Credential ID: JLX15LL45Z72	<i>Aug 2025</i>
Cloud and Generative AI Fundamentals on AWS – <i>Dicoding Indonesia</i> Credential ID: NVP75DYM4XR0	<i>May 2025</i>